NATIONAL
Junior College

SH2 Preliminary Examinations

8834 H1 Geography

Paper 1

Suggested Marking Guide

Disclaimer:
The Suggested Marking Guide should be treated with confidentiality, and should be used with discretion.

Notes on Marking

Marking Approaches

a. **Point marking** is used for marking all sub-parts of the structured questions, except for the 10-mark fieldwork evaluative sub-part.

   For point marking, the general rule is that each creditworthy response is awarded one mark. More detailed guidance is provided in the marking guide for each question.

b. **Generic level descriptors** are used for marking:

- 10-mark fieldwork evaluative sub-part structured questions; and
- 13-mark essays.

Possible approaches are provided for questions marked using generic level descriptors. They provide guidance on how the questions could be approached. The possible approaches provided are **neither exhaustive nor should they be treated as a model approach to questions**.

Generic Level Descriptors

Levels	Marks	Generic Level Descriptors for H1 Fieldwork Evaluative Question
3	9-10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context
2	5-8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1-4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Levels	Marks	Generic Level Descriptors for H1 Essays
4	11-13	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	8-10	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	5-7	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1-4	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Note:
Assessment using generic level descriptors involves qualitative rather than quantitative evaluation. Judgements on the level to be awarded to an answer will be based on the principle of 'best fit' determined by the descriptors within each level.

As a general guideline, responses deemed to have fulfilled all the descriptors within a level may be awarded the top mark in that level.

Section A

Cluster 2: Fieldwork

1		A group of eight students conducted a geographical investigation into the effectiveness of local food waste reduction efforts in the Bishan neighbourhood of Singapore. These efforts include a 24-hour community fridge and a food recycling locker at Block 112 Bishan Street 12, and periodic food rescue and redistribution events held at Bishan Community Club. Their hypothesis was: *"Residents who live nearer to community-based food waste initiatives are more likely to participate in them."* To test this hypothesis, the students first visited Block 112 Bishan Street 12 and Bishan Community Club on a Saturday afternoon, where they interviewed four resident volunteers to understand how the residents engage with local food waste initiatives. On the following Saturday, they then administered a questionnaire to 40 residents at Bishan Bus Interchange. Resource 1A is a map showing the locations of their investigation sites and food waste reduction efforts in the Bishan neighbourhood. Resource 1B shows photographs of food waste reduction efforts at Bishan Neighbourhood. Resource 2 shows selected interview responses. Resource 3 shows a survey form used by the students to collect responses. Resource 4 shows a summary of selected survey results.	
	(a)	Explain why this hypothesis is appropriate for investigating food waste reduction.	[4]
		Award 1 mark for each explanation of why the hypothesis is appropriate to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • [Target group] There is a **clearly defined population and variable** with the group of people and what is to be studied about them. • [Suitability of the scale] The hypothesis is of a **suitable spatial scale** of a neighbourhood. The hypothesis only investigates one neighbourhood in Singapore/Bishan. This is **manageable** for eight students to study over two days. • [Measurable] The hypothesis **links proximity** to **behaviour**. Resource 1A shows that both Block 112 and Bishan CC are important food rescue sites **within measurable walkable distances of residential sites in Bishan.**	

		• [Researchable] There are **eight students in this group**, which allows them to **split into two groups of four** to collect more responses at the two sites at same time. **AO1**	
	(b)	Suggest ethical considerations the group should take into account when collecting data on-site.	[3]
		Award 1 mark for each explanation point of ethical considerations the group should take to a maximum of 3 marks Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • **Ensure confidentiality of data collected. Ensure anonymity** and ensure that data collected including residential addresses will not be linked to personal identities / Inform respondents that their anonymity will be maintained and they will not be able to be identified. • **Obtain informed consent.** Participants at the bus interchange or community areas must agree to the survey and **understand its purpose**. • **Avoid disruption** of community activities. Food redistribution at Bishan Community Centre is a live event. Conducting of surveys **must not hinder volunteers** or recipients performing tasks **AO1**	
	(c)	With reference to Resource 2, describe how the data collected from the interviews can be analysed.	[5]
		Award 1 mark for each valid description up to maximum of 5 marks Award a maximum of 1 additional mark for further development of each description, where applicable. Possible answers include: • **Categorical coding of qualitative data** ○ This can be done to segment responses into themes such as motivation for participation / barriers for non-participation and behaviour as shown in Resource 2B (e.g. social bonding, privacy concerns, embarrassment, inconvenience) • **Comparison by demographic groups** ○ Students can compare difference in views of elderly vs younger residents as shown in Resource 2B • **Triangulation with digital data** ○ Students can incorporate screenshots of digital communications of app and contents on screenshots to validate self-reported participation (or lack of participation) as shown in Resource 2C **AO2**	

	(d)	With reference to Resource 3, explain the strengths of the questionnaire used in this investigation.	[4]
		Award 1 mark for each explanation point of the survey questions' strengths, up to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • The survey questions **cover a wide range of variables** including demographic, spatial, behavioural and attitudinal questions throughout Resource 3. • There is **clear differentiation between awareness of food waste reduction efforts and participation of them**. By asking separate questions for awareness (Q1) and participation (Q2), students can identify gaps between knowledge and behaviour • The multiple-choice format of questionnaire **enhances consistency** in responses. This **standardizes responses enabling easy-coding** and quantitative analysis. The inclusion of "Others: __" allows some qualitative input. • Questions **capture both motivations and barriers**. These provide balanced insight into participation decisions. Q3 for motivators and Q5 for barriers **AO2**	
	(e)	Using Resource 4, sketch a graph to present top reasons for participation in local food waste reduction efforts.	[4]
		Award 1 mark for each component of suitably sketched graph up to a maximum of 3 marks. Award a maximum of 1 mark for accuracy of graph. Possible components of graphical sketch include: • **Comparative** bar graph – bars must be to scale and spaced clearly • **Clear legend or labels** – reasons must be clearly annotated; distinguished by shading or colours • **Accurate plotting** – Environmental concern: 62.1%, Cost savings 48.3% and 37.9% for convenient location • **Title:** Top reasons for participation vs % of participation • **Axes labelled:** % of respondents – y-axis and Reasons for participation (environmental concern/cost savings/ convenient location – x-axis) **AO1**	

	(f)	Based on the students' investigation, the students concluded that their hypothesis was proven wrong. With reference to all resources, evaluate the validity of the fieldwork.	[10]
		Possible approaches: Candidates could consider the validity of their hypothesis **based on the data collected.** They could refer to Resources to see if there is **contradictory quantitative data**. Participants lived further from sites on average than non-participants (cite data from Resource 4) contradicting the hypothesis. Awareness levels and motivators like environmental concern or cost savings mattered more than proximity. Candidates could consider the way the investigation was conducted, including the data collection methods. They could **analyse the sequence of data collection methods** and suggest conducting survey questionnaires before carrying out interviews. The benefits of the improvised sequence could be examined. Candidates could also **analyse sample bias due to location**. All surveys were conducted at Bishan Bus Interchange potentially under-sampling non-users (Resources 3 & 4). Despite limitations, **multiple types of data (both quantitative questionnaire surveys and qualitative interviews** were carried out to strengthen conclusions. Interview quotes show age-related attitudes and social motivations affect participation as well. Awareness levels also mattered more than proximity. For the data collected in Resource 4, candidates could consider **the precision of data collection.** Distance measurements may lack precision – survey gave option for street, not exact unit limiting distance accuracy. There is **also lack of clarity of the terms used** in the questionnaire survey (e.g. occasionally/cost savings/environmental concern) and how these could give rise to differing interpretations by the residents. Candidates could consider how such **differing interpretations could affect the conclusions** that could be drawn from the analysis of these data. **Additional sources of secondary data** could also supplement the students' holistic understanding of the varied other factors that could influence local food waste reduction efforts besides proximity to location including data on food waste management from the National Environment Agency (NEA) and the Singapore Environment Council (SEC). *Levels marked using Generic Level Descriptors for H1 fieldwork evaluative questions* **AO3**	

Section B

Cluster 1: Sustainable Future and Climate Change

2	Resource 5 shows the change in liveability scores from 2021 to 2022 in the European region, from a Locations Ratings Survey done by ECA International, a company that provides data, software and global mobility expertise to multinational companies. Resource 6 shows an infographic on Copenhagen, Denmark as a bike-friendly city. Resource 7 shows 7 of the 17 Sustainable Development Goals (SDGs) indexes by the OECD's localised framework which normalises the SDG indicators from 0 to 100 – where 100 is the suggested end value of an indicator to be achieved by 2030.		
	(a)	Using Resource 5, describe the change in liveability scores from 2021 to 2022 in the European region.	[4]
		Award 1 mark for each description of the change in liveability scores from 2021 to 2022 in the European region up to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses: • In general, most cities saw a **decrease in liveability scores.** 14 out of the 22 cities identified had decreased liveability • Lisbon, Rome and Prague saw **the largest decrease** in liveability scores • However, 6 cities saw **an increase in liveability**. • Three cities saw **no change in liveability scores**- Copenhagen, Belgrade and Istanbul **AO2**	
	(b)	Explain the factors influencing the liveability of a place.	[5]
		Award 1 mark for each explanation of factor influencing the liveability of a place up to maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. <u>Possible responses</u> • The liveability of a place is **influenced by its climate**. In general, it is agreed that mild temperatures and low risks of weather-related disasters help to make a place more liveable. • Another environmental factor influencing the liveability of a place is the **quality of its environment.** • **The quality of air, water and land in cities** are important components of liveability for both health and aesthetics reasons. • The **access to services and facilities** helps make a place more liveable as the essential needs of people are better met, and the quality of living is higher as well	

		• Social factors influencing the liveability of a place include **safety and stability** too. In general, cities are more liveable when a city has a **low crime rate, and is politically stable** (free from conflict) **AO1**	
	(c)	Describe how strategies such as those implemented by Copenhagen shown in Resource 6 can help cities respond to contemporary climate change.	[4]
		Award 1 mark for each description of strategy and how it helps cities respond to climate change up to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • **The promotion of cycling reduces greenhouse gas emissions** – 92% less carbon dioxide emissions when changing from car to bike • **Investments toward the green agenda** can bring about socioeconomic benefits - Investments of 295 million euros for 746km of cycle superhighways by 2045. • **Green transport policies improve urban microclimates.** • **Shifting commuter habits supports the strengthening of community and social resilience** **AO2**	
	(d)	Using Resource 7, compare the sustainability performance between Copenhagen, Denmark and Zurich, Switzerland.	[4]
		Award 1 mark for each similarity or difference in sustainability performance between Copenhagen and Zurich. Award a maximum of 1 additional mark for further development of each similarity or difference. Award a maximum of 1 mark if only similarities or differences are addressed. Possible answers include: **Differences** • Zurich outperforms Copenhagen in three SDG goals (6, 11 and 16) • Copenhagen outperforms Zurich in three SDG goals (SDG 5, 13 and 15) **Similarities** • Both Zurich and Copenhagen achieved maximum 100 points for SDG 17 • Both Zurich and Copenhagen performed relatively well for SDG 11, 16 and 17 with above 80 points for each SDG goal. • Both Zurich and Copenhagen fared relatively poorly for SDG 15, with Copenhagen scoring 56 points and Zurich scoring 33 points. **AO2**	

	(e)	With reference to Resources 6 and 7, suggest possible challenges in progressing towards sustainable development.	[5]
		Award 1 mark for each suggestion of possible challenges in progressing towards sustainable development up to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable. Possible answers include: • SDG 6 (Resource 7) - **Urban demand strains water resources**. • SDG 11 (Resource 7) and Bike-friendly Cities (Resource 6) - **Energy consumption in cities** impacts progress. Urbanisation increases carbon emissions o **Urban sprawl** (in cities such as Zurich) strain public transport systems and **undermine inclusivity** in urban living. • **Protection of forested areas** (SDG 15)– increasing proportion of protected areas for **terrestrial biodiversity** o Rapid urbanisation consumes surrounding forests, leading to habitat fragmentation. This threatens biodiversity and reduces ecosystem services like carbon storage and increases vulnerability to flooding. • SDG 16 (Resource 7) – Both cities enjoy trust in **government and low crime rates**. However, there are **concerns over inclusiveness and integration of migrants**. Copenhagen has seen political debates around ghetto policies and housing segregation **AO1**	
3		Resource 8A shows atmospheric carbon dioxide levels from 800,000 years ago based on ice core data to current levels in 2025. Resource 8B shows global ocean heat content trends from 1955 to 2020. Resource 9 shows cloud feedback mechanisms.	
	(a)	Using Resource 8A, describe how atmospheric carbon dioxide levels have changed over the past 800,000 years.	[4]
		Award 1m for each description of the trend to a maximum of 2 marks. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses include: • **Carbon dioxide levels have fluctuated cyclically for most of the past 800,000 years** • Significant increase within the most recent centuries. • **A sharp dramatic rise in carbon dioxide levels continued from 1911 to 1958**. Current levels are unprecedented throughout the 800,000-year history • The **amount of carbon dioxide has increased to over 420 ppm currently.** **AO2**	

	(b)	Explain how ice cores serve as proxy indicators of past climate variability.	[4]
		Award 1 mark for each explanation of ice cores as proxy indicator. Award a maximum of 1 additional mark for further development of each description, where applicable. Possible responses include: • Analysis of ice cores show **evidence of a regular cycle of warming and cooling of the Earth's climate in the past**. **Isotope analysis of ice core oxygen content** indicates that there were warmer interglacial and cooler glacial intervals • Analysis of ice cores shows that a **higher concentration of greenhouse gases** coincided with periods of higher global temperatures - the examination of the gases trapped in ice cores have shown scientists that the amount of carbon dioxide and the global temperature have been linked throughout at least in the last million years of Earth's history. **AO1**	
	(c)	With reference to 8A and 8B, explain the significance of human activities on the global carbon cycle.	[5]
		Award 1 mark for explanation of the magnitude and rate of change in carbon dioxide levels in the atmosphere as well as oceans and linking these to human activity, up to a maximum of Award a maximum of 1 additional mark for further development of each suggestion where applicable. Possible responses include: • **Rapid increase in atmospheric carbon dioxide due to human activities** • (Resource 8A) shows that atmospheric carbon dioxide concentrations remained below 300 ppm over the past 800,000 years but role sharply post-industrialisation after 1911. • **Oceans as a major carbon sink** (Resource 8B) shows a steady increase in global ocean heat content from 1955 to 2020, which is evidence of excess heat being absorbed by oceans due to elevated carbon dioxide levels. • As more carbon dioxide accumulates in the atmosphere from human sources, oceans absorb both carbon and heat, weakening their ability to buffer climate change over time **AO1**	
	(d)	Explain the impact of contemporary climate change on aquatic ecosystems.	[4]
		Award 1 mark for each explanation of impact up to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation. Possible responses include:	

		• **Deoxygenation of oceans with warmer waters** being able to hold less oxygen leading to hypoxic conditions. Low oxygen levels reduce survival of fish and invertebrates, especially in coastal waters, impacting ecosystem services and commercial fishing • **Changes in migration and breeding patterns** – warming waters shift species distributions and migration timing. Fish stocks may move poleward or to deeper waters disrupting ecosystems and affecting fisheries and food security • **Ocean warming and Coral bleaching** - Rising sea temperatures stress coral reefs leading to coral bleaching. Corals expel symbiotic algae, leading to widespread bleaching events and loss of biodiversity • **Ocean acidification and marine life**- Increased carbon dioxide absorption by oceans lower pH, causing ocean acidification. This reduces the ability of calcifying organism such as shellfish, corals and plankton to form shells and skeletons, disrupting marine food webs **AO1**	
	(e)	With reference to Resource 9, explain how clouds in a warming climate illustrate uncertainty over the future impact of contemporary climate change.	[5]
		Award 1 mark for each explanation of feedback mechanism, up to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each explanation. Possible responses include: • **Clouds can amplify climate change through positive feedback.** Positive feedback occurs when warming causes cloud changes that further enhance warming. • If global warming leads to a **reduction in low-lying clouds over oceans**, more solar energy is absorbed by the Earth's surface, raising temperatures even more. • This **reinforces the warming trend**, creating a **loop that accelerates** climate change and global warming. • However **negative feedback mechanisms can occur when changes in cloud properties mitigate the effects of global warming**. • An **increase in high-albedo cloud cover** might reflect more incoming solar radiation back into space, thereby cooling the Earth's surface. • **Low-lying clouds reflect incoming solar radiation**, creating negative feedback that cools the planet • **Limitations in satellite data** showing the effects of clouds in a warming climate exacerbates the uncertainty. Current levels of technology and **global climate models** do not fully capture the true extent of future impact. **AO2**	

Section C

Cluster 1: Sustainable Future and Climate Change

4	"Community-led initiatives are more effective than government-led programmes in upgrading slums." Evaluate this statement.	[13]
	Possible Approaches: Candidates could approach the question by making a judgement on whether community-led initiatives or government-led initiatives are more effective in upgrading slums. Candidates **could analyse success criteria of various strategies by weighing the strengths and limitations of both community and government approaches to slum upgrading, and by drawing from a variety of case studies to support a reasoned argument.** Candidates may examine if initiative target the root causes of locals' needs (e.g. community led initiatives such as Orangi Pilot Project, Pakistan which often respond to residents' specific priorities) and compare government programmes that may adopt a top-down model that may not align with community needs. Candidates may also assess the scalability and resources of governments which may typically have greater financial, legal and administrative capacity to scale-up projects and integrate slum upgrading into national development. Candidates could also approach the question by **evaluating context-specific outcomes using case studies of public housing, site and services scheme as well as self-help upgrading schemes. Candidates could consider the effectiveness of the initiatives through examining two or more case studies that illustrate how effectiveness vary depending on political, social and spatial context.** Candidates could evaluate the political will and institutional capacity between weak and responsive governance, the provision of land titles or tenure, or provide case studies where there is support from NGOs and international donors and collaboration across stakeholders. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	

5	Evaluate the extent to which cities have successfully met the needs of women.	[13]
	Possible Approaches: Candidates could approach the question by making a judgement on whether strategies to address issues faced by women have been largely successful through a **consideration of case studies put in place by urban authorities to cater to the needs of women, bearing in mind intersectionalities. Candidates could evaluate whether the strategies in each case study achieved the intended aims to cater to the economic, social and psychological well-being of different groups of women.** Candidates could also analyse the issues faced by women which are common across the case studies including issues pertaining to gender equality, employment and gender pay gap, burden of unpaid care work, access to maternity leave and impact on their physical and mental health, transport, political representation and participatory processes in urban planning. Candidates could also **evaluate the challenges to success across these case studies** including the lack of access to essential needs, discrimination, as well as spatial and social exclusion. Candidates could also approach the question by making a judgement on whether strategies to **address issues faced by women have been successful through a cost-benefit analysis. Candidates could analyse the positive and negative impacts of the strategies in terms of economic scale, temporal scale and impacts on progress towards sustainable development.** Candidates could examine the extent to which the strategies meet the diverse range of needs of women and the extent to which various stakeholders are able to implement these strategies successfully. Candidates could also analyse the level of success achieved to meet urgent short-term needs of the women vis-à-vis the long-term challenges of developing a sustainably gender-inclusive city. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	

6	"The impact of climate change is not equal across places and populations." Evaluate this statement.	[13]
	Candidates could approach the question by making a judgement on whether contemporary climate change will affect everyone equally. They may choose to do this through a **consideration of criteria such as the scale, extent and irreversibility of impacts. Candidates could explain how the differences in the extent of exposure to the effects that each region has, as well as the extent of vulnerability caused by variations in the levels of development and technological expertise, and degree of preparedness.** Candidates may analyse how the impacts of climate change may vary due to differences in physical conditions (such as the uneven changes in temperature and precipitation across places) and vulnerabilities of different groups of people (such as uneven vulnerabilities across different ages, gender, and income groups) and varied levels of adaptive capacity. Candidates could also approach the question by making on judgement on whether contemporary climate change will affect everyone equally through a consideration of two or more case studies. Candidates could analyse the contextual factors that may contribute towards variations in the impact of contemporary climate change. Responses should include a **discussion of the different effects of climate change such as rising sea levels and its impact on agriculture and food security, as well as the spread of infectious diseases and heat-related mortality, and how these compare across developed and developing regions.** There should also be an understanding of how the challenges faced between regions of different levels of development influence their ability to adapt and to mitigate the impacts of climate change. *Levels marked using Generic Level Descriptors for 13m H1 essays* **AO3**	